

Digit and Numbers

LEVEL 1: Digit Extraction Basics

Program 1: Extract Last Digit Write a program that takes any number as input and extracts and displays the last digit of that number.

Program 2: Remove Last Digit Write a program that takes a number as input and displays the number after removing its last digit.

Program 3: Extract First Digit of 3-Digit Number Write a program that takes a 3-digit number as input and extracts and displays the first digit.

Program 4: Extract Middle Digit of 3-Digit Number Write a program that takes a 3-digit number as input and extracts and displays the middle digit.

Program 5: Sum of Digits (3-digit number) Write a program that takes a 3-digit number as input and calculates the sum of all its digits, then displays the result.

Program 6: Reverse a 2-Digit Number Write a program that takes a 2-digit number as input and displays it in reverse order (e.g., 45 becomes 54).

Program 7: Reverse a 3-Digit Number Write a program that takes a 3-digit number as input and displays it in reverse order (e.g., 456 becomes 654).

Program 8: Swap First and Last Digit of 3-Digit Number Write a program that takes a 3-digit number as input and swaps its first and last digits, then displays the result (e.g., 456 becomes 654).

Program 9: Average of First and Last Digit Write a program that takes a 4-digit number and calculates the average of its first and last digits.

Program 10: Product of All Digits Write a program that takes a 3-digit number as input and calculates the product of all its digits, then displays the result.

LEVEL 2: Number Properties

Program 11: Check if Last Digit is Even Write a program that takes any number as input and checks whether its last digit is even or odd, then displays the result.

Program 12: Palindrome Number Checker (2-digit) Write a program that takes a 2-digit number as input and checks if it is a palindrome (reads same forwards and backwards). Example: 11 → Yes, 23 → No. Display "Palindrome" or "Not Palindrome".

Program 13: Palindrome Number Checker (3-digit) Write a program that takes a 3-digit number as input and checks if it is a palindrome. Example: 121 → Yes, 123 → No. Display "Palindrome" or "Not Palindrome".

Program 14: Find Largest Digit in a Number Write a program that takes a 4-digit number as input and finds the largest digit in it, then displays that digit.

Program 15: Find Smallest Digit in a Number Write a program that takes a 4-digit number as input and finds the smallest digit in it, then displays that digit.

Program 16: Check Divisibility by 3 Using Digit Sum Write a program that takes a number as input, calculates the sum of its digits, and checks if the number is divisible by 3 using the digit sum rule.

Program 17: Count Even Digits in a Number Write a program that takes a 4-digit number as input and counts how many of its digits are even, then displays the count.

Program 18: Check if Digits are in Ascending Order Write a program that takes a 3-digit number as input and checks if its digits are in strictly ascending order (e.g., 123 is yes, 132 is no).

Program 19: Harshad/Niven Number Checker (3-digit) Write a program that takes a 3-digit number, calculates the sum of its digits, and checks if the number is divisible by this sum. Example: 153 → $1+5+3=9$, $153 \div 9=17$ → Harshad Number. Display "Harshad Number" or "Not Harshad Number".

Program 20: Duck Number Checker (4-digit) Write a program that takes a 4-digit number ABCD and checks if it contains the digit 0 anywhere

except the first position. Example: 4012 → Duck Number, 0123 → Not Duck, 1234 → Not Duck. Display "Duck Number" or "Not Duck Number".

Program 21: Spy Number Checker (3-digit) Write a program that takes a 3-digit number, calculates both the sum of its digits and the product of its digits, and checks if both are equal. Example: 123 → sum=6, product=6 → Spy Number. Display "Spy Number" or "Not Spy Number".

Program 22: Bouncy Number Checker (3-digit) Write a program that takes a 3-digit number ABC and checks if its digits are neither in strictly ascending order nor in strictly descending order. Example: 132 → Bouncy, 123 → Not Bouncy. Display "Bouncy" or "Not Bouncy".

Program 23: Neon Number Checker (Single Digit) Write a program that takes a single digit number (0-9), calculates its square, finds the sum of digits of the square, and checks if this sum equals the original number. Example: 9 → $9^2=81$ → $8+1=9$ → Neon. Display "Neon Number" or "Not Neon Number".

Program 24: Perfect Square Checker (2-digit) Write a program that takes a 2-digit number and checks if it is a perfect square by comparing against known values: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100. Display "Perfect Square" or "Not Perfect Square".

LEVEL 3: Advanced Number Patterns

Program 25: Armstrong Number Checker (3-digit) Write a program that takes a 3-digit number ABC and checks if $A^3 + B^3 + C^3$ equals the original number. Example: 153 → $1^3+5^3+3^3 = 1+125+27 = 153$ → Armstrong Number. Display "Armstrong Number" or "Not Armstrong Number".

Program 26: Digital Root Calculator Write a program that takes a number as input and finds its digital root by repeatedly adding its digits until a single digit remains (e.g., 9875 → $9+8+7+5=29$ → $2+9=11$ → $1+1=2$).

Program 27: Narcissistic Number Checker Write a program that takes a 3-digit number, calculates sum of cubes of its digits: $(d_1^3 + d_2^3 + d_3^3)$. Display both original number and the calculated sum (for comparison).

Program 28: Magic Number Generator Write a program that takes a 2-digit number AB, generates a 4-digit magic number using pattern: $(A \times 10 + B) \times 101$. Example: 23 $\rightarrow$ 2323. Display the magic number.

Program 29: Digit Frequency Analyzer Write a program that takes a 5-digit number and counts occurrences of specific digits: count how many 0's, how many 5's, how many 9's. Display three counts.

Program 30: Kaprekar Routine (Single Step) Write a program that takes a 4-digit number, extracts all digits, manually arrange to form largest possible number and smallest possible number, then subtract. Display the result. Example: 3524 $\rightarrow$ largest: 5432, smallest: 2345 $\rightarrow$ 5432-2345 = 3087

Program 31: Automorphic Number Checker (2-digit) Write a program that takes a 2-digit number, calculates its square, extracts the last 2 digits of the square, and checks if they match the original number. Example: 25 $\rightarrow$ $25^2=625 \rightarrow$ last 2 digits=25 $\rightarrow$ Automorphic. Display "Automorphic" or "Not Automorphic".

Program 32: Trimorphic Number Checker (2-digit) Write a program that takes a 2-digit number, calculates its cube, extracts the last 2 digits of the cube, and checks if they match the original number. Example: 24 $\rightarrow$ $24^3=13824 \rightarrow$ last 2 digits=24 $\rightarrow$ Trimorphic. Display "Trimorphic" or "Not Trimorphic".

Program 33: Sunny Number Checker Write a program that takes a number N, calculates $N+1$, and checks if $N+1$ is a perfect square by comparing with known perfect squares (1, 4, 9, 16, 25, 36, 49, 64, 81, 100). Example: 8 $\rightarrow$ $8+1=9 \rightarrow$ Sunny Number. Display "Sunny Number" or "Not Sunny Number".

Program 34: Pronic Number Checker (2-digit) Write a program that takes a 2-digit number and checks if it can be expressed as the product of two consecutive numbers $n \times (n+1)$. Check against: 2, 6, 12, 20, 30, 42, 56, 72, 90. Example: 12 $\rightarrow$ $3 \times 4=12 \rightarrow$ Pronic. Display "Pronic" or "Not Pronic".

Program 35: Disarium Number Checker (3-digit) Write a program that takes a 3-digit number ABC and checks if $A^1 + B^2 + C^3$ equals the original

number. Example: $175 \rightarrow 1^1 + 7^2 + 5^3 = 1 + 49 + 125 = 175 \rightarrow$ Disarium. Display "Disarium Number" or "Not Disarium Number".

Program 36: Tech Number Checker (4-digit) Write a program that takes a 4-digit number ABCD, splits it into two 2-digit numbers AB and CD, calculates $(AB + CD)^2$, and checks if the result equals ABCD. Example: $2025 \rightarrow 20 + 25 = 45 \rightarrow 45^2 = 2025 \rightarrow$ Tech Number. Display "Tech Number" or "Not Tech Number".

Program 37: Kaprekar Number Checker (2-digit) Write a program that takes a 2-digit number, calculates its square, splits the square into two parts, adds these parts, and checks if the sum equals the original number. Example: $45 \rightarrow 45^2 = 2025 \rightarrow 20 + 25 = 45 \rightarrow$ Kaprekar. Display "Kaprekar Number" or "Not Kaprekar Number".

Program 38: Gapful Number Checker (3-digit) Write a program that takes a 3-digit number ABC, forms a 2-digit number using the first and last digits ($10 \times A + C$), and checks if ABC is divisible by this 2-digit number. Example: $132 \rightarrow 12 \rightarrow 132 \div 12 = 11 \rightarrow$ Gapful. Display "Gapful Number" or "Not Gapful Number".